1) $$\int \frac{2e^{\operatorname{sen}(x)}\left(x\cos^3(x)-\operatorname{sen}(x)\right)}{1+\cos(2x)}\,dx$$

⇨ Solución:

→ $$I=\int \frac{\cancel{2}\,e^{\operatorname{sen}(x)}\left(x\cos^3(x)-\operatorname{sen}(x)\right)}{\cancel{2}\cos^2(x)}\,dx$$

$$I=\int x\,e^{\operatorname{sen}(x)}\cos(x)\,dx-\int \frac{e^{\operatorname{sen}(x)}\operatorname{sen}(x)}{\cos^2(x)}\,dx$$

→ Por partes:

$$\left.\begin{array}{l} u=x \\ du=dx \end{array}\right\}\begin{array}{l} \int dv=\int e^{\operatorname{sen}(x)}\cos(x)\,dx \\ v=e^{\operatorname{sen}(x)} \end{array}$$

$$\left.\begin{array}{l} K=e^{\operatorname{sen}(x)} \\ dK=e^{\operatorname{sen}(x)}\cos(x) \end{array}\right\}\begin{array}{l} \int dP=-\int \frac{\operatorname{sen}(x)}{\cos^2(x)}\,dx \\ P=-\frac{1}{\cos(x)}=-\sec(x) \end{array}$$

→ $$I=x\,e^{\operatorname{sen}(x)}-\int e^{\operatorname{sen}(x)}\,dx-e^{\operatorname{sen}(x)}\sec(x)+\int \frac{1}{\cancel{\cos(x)}}\cdot e^{\operatorname{sen}(x)}\cancel{\cos(x)}\,dx$$

$$I=e^{\operatorname{sen}(x)}\left(x-\sec(x)\right)-\cancel{\int e^{\operatorname{sen}(x)}\,dx}+\cancel{\int e^{\operatorname{sen}(x)}\,dx}$$

→ $$\boxed{I=e^{\operatorname{sen}(x)}\left(x-\sec(x)\right)+C}$$

(2) $$\int \frac{x^4+8x^3+30x^2+53x+38}{(x^2+4x+6)^2(x+2)}\,dx$$

⇨ Solución:

$$\to I = \int \frac{(x^4+8x^3+28x^2+48x+36)+(2x^2+5x+2)}{(x^4+8x^3+28x^2+48x+36)(x+2)}$$

$$I = \int \frac{1}{x+2}\,dx + \int \frac{(2x+1)\cancel{(x+2)}}{(x^2+4x+6)^2\cancel{(x+2)}}\,dx$$

$$I = \ln|x+2| + \underbrace{\int \frac{2x+1}{((x+2)^2+2)^2}\,dx}_{II}$$

Sustitución:

$u = x+2 \to x = u-2$

$du = dx$

$$\to II = \int \frac{2(u-2)+1}{(u^2+2)^2}\,du = \int \frac{2u-3}{(u^2+2)^2}\,du = \int \frac{2u}{(u^2+2)^2}\,du - 3\int \frac{1}{(u^2+2)^2}\,du$$

↳ $t = u^2+2$

$dt = 2u\,du$

↳ Integral Directa de la forma:

$$\int \frac{1}{(K^2+a^2)^2}\,dK$$

$$\to I = \ln|x+2| + \int \frac{1}{t^2}\,dt - 3\left(\frac{1}{2(2)}\right)\left[\frac{u}{u^2+2} + \frac{1}{\sqrt{2}}\arctan\left(\frac{u}{\sqrt{2}}\right)\right] + C$$

↳ Devolvemos Sustitución:

$$\to \boxed{I = \ln|x+2| - \frac{3x+10}{4(x^2+4x+6)} - \frac{3}{4\sqrt{2}}\arctan\left(\frac{x+2}{\sqrt{2}}\right) + C}$$

③ $$\int \frac{3x^2+4}{2\sqrt{x}\,(4-3x^2)\sqrt{3x^2+x-4}}\,dx$$

⇨ Solución:

$$\to I = \int \frac{3x^2+4}{-2\sqrt{x}\,(3x^2-4)\sqrt{x}\sqrt{3x+1-\frac{4}{x}}}\,dx$$

$$I = -\int \frac{3x^2+4}{2x^2\left(3x-\frac{4}{x}\right)\sqrt{3x-\frac{4}{x}+1}}\,dx$$

$$I = -\int \frac{3+\frac{4}{x^2}}{2\left(3x-\frac{4}{x}\right)\sqrt{3x-\frac{4}{x}+1}}\,dx \quad \Bigg\} \begin{array}{l} \text{Sustitución:} \\ u = 3x-\frac{4}{x} \\ du = \left(3+\frac{4}{x^2}\right)dx \end{array}$$

$$\to I = -\int \frac{1}{2u\sqrt{u+1}}\,du \quad \Bigg\} \begin{array}{l} \text{Sustitución:} \\ t^2 = u+1 \to u = t^2-1 \\ 2t\,dt = du \end{array}$$

$$\to I = -\int \frac{\cancel{2t}}{\cancel{2}(t^2-1)\cancel{t}}\,dt = -\int \frac{1}{t^2-1}\,dt = \int \frac{1}{1-t^2}\,dt$$

↳ Integral Directa

$$\to I = \operatorname{arctanh}(t) + C$$

Devolvemos Sustituciones

$$\to \boxed{I = \operatorname{arctanh}\left(\sqrt{\frac{3x^2-4+x}{x}}\right) + C}$$

(4) $$\int \frac{\operatorname{sen}(x)+x\cos(x)}{(1-x\operatorname{sen}(x))\sqrt{-1+x^2-x^2\cos^2(x)}}\,dx$$

⇨ Solución:

→ Sustitución:

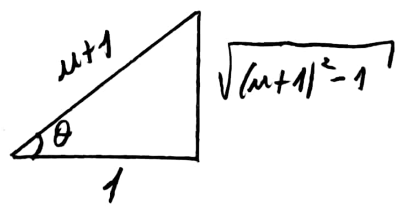
$$u=x\operatorname{sen}(x)-1 \longrightarrow x\operatorname{sen}(x)=u+1$$

$$du=(\operatorname{sen}(x)+x\cos(x))\,dx \qquad x^2\operatorname{sen}^2(x)=(u+1)^2$$

$$x^2\operatorname{sen}^2(x)-1=(u+1)^2-1$$

→ $$-\int \frac{1}{(x\operatorname{sen}(x)-1)\sqrt{x^2(\underbrace{1-\cos^2(x)}_{\to\ \operatorname{sen}^2(x)})-1}}\,du$$

→ $$I=-\int \frac{1}{u\sqrt{(u+1)^2-1}}\,du$$



$\to u+1=\sec(\theta) \longrightarrow u=\sec(\theta)-1$

$du=\sec(\theta)\tan(\theta)\,d\theta$

$\to \sqrt{(u+1)^2-1}=\tan(\theta)$

→ $$I=-\int \frac{1}{(\sec(\theta)-1)\cancel{\tan(\theta)}}\cdot \sec(\theta)\cancel{\tan(\theta)}\,d\theta=-\int\frac{\sec(\theta)}{\sec(\theta)-1}\,d\theta$$

→ $$I=-\int \frac{\sec(\theta)(\sec(\theta)+1)}{\tan^2(\theta)}\,d\theta=-\int\left(\csc^2(\theta)+\frac{\cos(\theta)}{\operatorname{sen}^2(\theta)}\right)d\theta$$

$$I=-\left(-\operatorname{ctg}(\theta)-\frac{1}{\operatorname{sen}(\theta)}\right)+C=\frac{1}{\sqrt{(u+1)^2-1}}+\frac{u+1}{\sqrt{(u+1)^2-1}}+C$$

Devolvemos Sustitución

→ $$\boxed{I=\frac{1+x\operatorname{sen}(x)}{\sqrt{x^2\operatorname{sen}^2(x)-1}}+C}$$

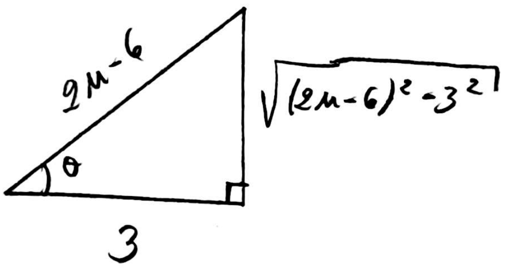
(5) $$\int \frac{\cosh}{(4\sinh^2(x)-24\sinh(x)+27)\sqrt{4\sinh^2(x)-24\sinh(x)+27}}\,dx$$

⇨ Solución:

↳ Sustitución:

$$u=\sinh(x)$$
$$du=\cosh(x)\,dx$$

↳ $$I=\int\frac{1}{(4u^2-24u+27)^{3/2}}\,du=\int\frac{1}{[(2u-6)^2-3^2]^{3/2}}\,du$$



$$\to 2u-6=3\sec(\theta)$$
$$2\,du=3\sec(\theta)\tan(\theta)\,d\theta$$
$$du=\frac{3}{2}\sec(\theta)\tan(\theta)\,d\theta$$

$$\to \sqrt{(2u-6)^2-3^2}=3\tan(\theta)$$

↳ $$I=\int\frac{1}{(3\tan(\theta))^3}\cdot\frac{3}{2}\sec(\theta)\tan(\theta)\,d\theta$$

$$I=\frac{1}{18}\int\frac{\cos(\theta)}{\sin^2(\theta)}\,d\theta=\frac{-1}{18\sin(\theta)}+C$$

↳ $$I=\frac{-1}{18}\left[\frac{2u-6}{\sqrt{(2u-6)^2-3^2}}\right]+C$$

Devolvemos Sustitución

↳ $$\boxed{I=\frac{3-\sinh(x)}{9\sqrt{4\sinh^2(x)-24\sinh(x)+27}}+C}$$

6 $\int \cos^n(x)\,dx$

⇨ Solución:

$$I = \int \cos^n(x)\,dx = \int \cos^{n-1}(x)\cos(x)\,dx$$

Por partes:

$$\left.\begin{aligned} u &= \cos^{n-1}(x) \\ du &= -(n-1)\cos^{n-2}(x)\,\mathrm{sen}(x)\,dx \end{aligned}\right\} \quad \begin{aligned} \int dv &= \int \cos(x)\,dx \\ v &= \mathrm{sen}(x) \end{aligned}$$

$$I = \mathrm{sen}(x)\cos^{n-1}(x) - \int \mathrm{sen}(x)\cdot -(n-1)\cos^{n-2}(x)\,\mathrm{sen}(x)\,dx$$

$$I = \mathrm{sen}(x)\cos^{n-1}(x) + \int (1-\cos^2(x))\,(n-1)\cos^{n-2}(x)\,dx$$

$$I = \mathrm{sen}(x)\cos^{n-1}(x) + (n-1)\int \cos^{n-2}(x)\,dx - \underbrace{\int \cos^n(x)\,dx}_{I}\cdot(n-1)$$

$$I(1+n-1) = I(n)$$

$$\boxed{I = \frac{\mathrm{sen}(x)\cos^{n-1}(x)}{n} + \frac{n-1}{n}\int \cos^{n-2}(x)\,dx}$$

⑦ $$\lim_{n\to\infty}\sum_{i=1}^{n}\frac{n}{(3n+i)\sqrt{27n^2+24in+4i^2}}$$

⇨ <u>Solución:</u>

↳ $$\lim_{n\to\infty}\sum_{i=1}^{n}\frac{n}{(3n+i)\left(\frac{n}{n}\right)\sqrt{n^2\left(27+24\frac{i}{n}+4\frac{i^2}{n^2}\right)}}$$

$$\lim_{n\to\infty}\sum_{i=1}^{n}\frac{1}{\left(3+\frac{i}{n}\right)\sqrt{27+24\frac{i}{n}+4\frac{i^2}{n^2}}}\cdot\frac{1}{n}$$

↳ $\Delta x=\frac{b-a}{n}\rightarrow\frac{1}{n}=\frac{1-0}{n}$

↳ $\boxed{a=0}$; $\boxed{b=1}$

Para : $f(Inf+i\Delta x)=f\left(0+i\left(\frac{1}{n}\right)\right)$

$=f\left(\frac{i}{n}\right)$ ✓ Concuerda

↳ $x_i=\frac{i}{n}$

↳ <u>Tenemos:</u>

$$\lim_{n\to\infty}\sum_{i=1}^{n}\frac{1}{(3+x_i)\sqrt{27+24x_i+4x_i^2}}\cdot\Delta x$$

↳ $$I=\lim_{n\to\infty}\sum_{i=1}^{n}f(x_i)\,\Delta x$$

↳ $$\boxed{I=\int_0^1\frac{1}{(3+x)\sqrt{27+24x+4x^2}}\,dx}$$

⑧ $y = x^3 + 2x - 3$, $x = 0$, $x = 3$, eje x.

Hallar área por definición.

⇨ Solución:

→ Intersección:

$y = 0$

↳ $x^3 + 2x - 3 = 0$

$(x-1)(x^2 + x + 3) = 0$

$\boxed{x = 1}$

$x = 0$

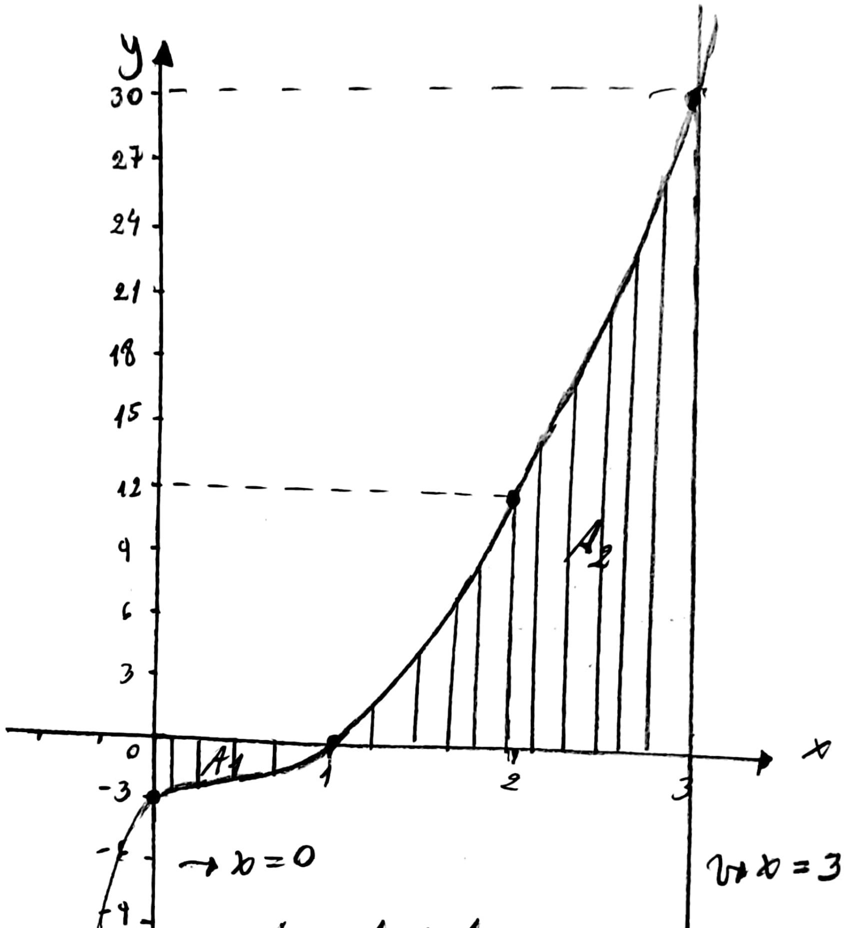
↳ $y = (0)^3 + 2(0) - 3$

$\boxed{y = -3}$

$x = 3$

↳ $y = (3)^3 + 2(3) - 3$

$\boxed{y = 30}$



$A_T = A_1 + A_2$

→ Áreas: $A_1 = \int_0^1 -(x^3 + 2x - 3)\,dx$; $A_2 = \int_1^3 (x^3 + 2x - 3)\,dx$

$$* \; A_1 = -\left(\frac{1-0}{n}\right) \lim_{n \to \infty} \sum_{K=1}^{n} \left[f\left(0 + \left(\frac{1-0}{n}\right) K\right)\right]$$

$$A_1 = \frac{-1}{n} \lim_{n \to \infty} \sum_{K=1}^{n} \left[f\left(\frac{K}{n}\right)\right] = \frac{-1}{n} \lim_{n \to \infty} \sum_{K=1}^{n} \left[\left(\frac{K}{n}\right)^3 + 2\left(\frac{K}{n}\right) - 3 \right]$$

$$A_1 = -\lim_{n \to \infty} \left[\frac{1}{n^4} \left(\frac{n(n+1)}{2}\right)^2 + \frac{2}{n^2}\left(\frac{n(n+1)}{2}\right) - \frac{3}{n}(n) \right]$$

$$A_1 = -\left[\frac{1}{4} + 1 - 3\right] = -\left[\frac{1-8}{4}\right] \to \boxed{A_1 = \frac{7}{4}}$$

⑧ 2da parte:

$$* A_2 = \frac{3-1}{n} \lim_{n\to\infty} \sum_{K=1}^{n} \left[f\left(1+\left(\frac{3-1}{n}\right)K\right)\right]$$

$$A_2 = \frac{2}{n} \lim_{n\to\infty} \sum_{K=1}^{n} \left[\left(1+\frac{2}{n}K\right)^3 + 2\left(1+\frac{2}{n}K\right) - 3\right]$$

$$A_2 = 2 \lim_{n\to\infty} \sum_{K=1}^{n} \left[\frac{10}{n^2}K + \frac{12}{n^3}K^2 + \frac{8}{n^4}K^3\right]$$

$$A_2 = 2 \lim_{n\to\infty} \left[\frac{10}{n^2}\left(\frac{n(n+1)}{2}\right) + \frac{12}{n^3}\left(\frac{n(n+1)(2n+1)}{6}\right) + \frac{8}{n^4}\left(\frac{n(n+1)}{2}\right)^2\right]$$

$$A_2 = 2\left(\frac{10}{2} + \frac{12(2)}{6} + \frac{8}{4}\right) = 2(5+4+2) = 2(11)$$

$$\boxed{A_2 = 22}$$

↪ Finalmente:

$$A_T = A_1 + A_2$$

$$A_T = \frac{7}{4} + 22 = \frac{7+88}{4}$$

$$\hookrightarrow \boxed{A_T = \frac{95}{4}\, m^2}$$